

From Raw Data to Business Decisions



A four-part data analytics internship journey across data cleaning, business intelligence, deep-dive analysis and statistically validated recommendations.



Aakrti

Four Tasks, One Analytical Journey

The internship progressed from data quality to business decisions, with each task adding a deeper layer of analysis.

01 CLEAN

Task 1

1,005 raw sales records inspected; missing values, duplicates and date issues addressed.

995 records retained

02 EXPLORE

Task 2

SQL cleaning + Excel BI dashboard used to answer 7 core business questions.

1,500 orders analyzed

03 DEEP-DIVE

Task 3

Python + Tableau used to define KPIs and examine sales versus profit by category and segment.

5,111 orders · 4 yrs

04 VALIDATE

Task 4

A profitability observation was converted into a hypothesis and tested statistically.

$p < 0.001$

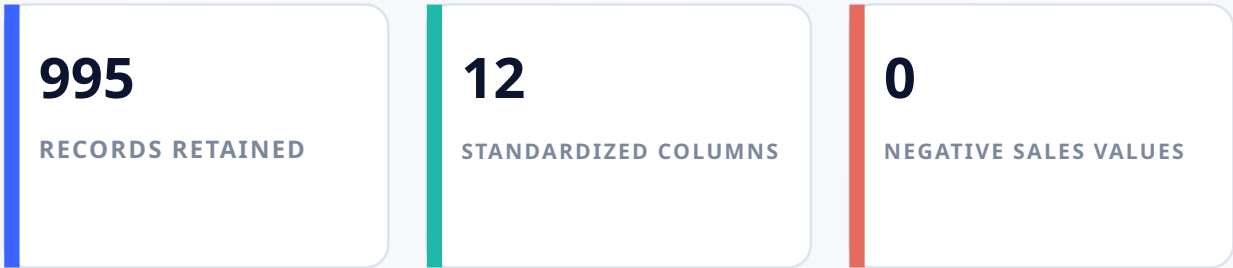
Building a Reliable Base

Sales_Dataset_1005.csv → cleaned with Python & Pandas

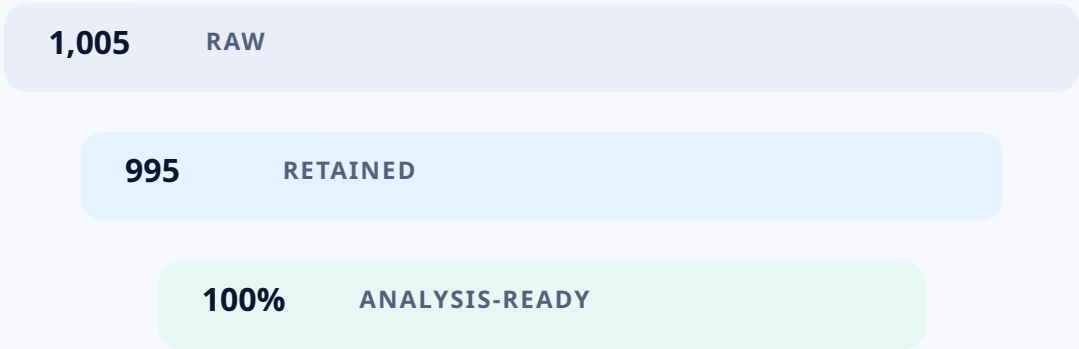
ISSUES IDENTIFIED

- 1 Missing Age values
- 2 Blank customer names
- 3 Duplicate transaction records
- 4 Negative (invalid) sales values
- 5 Inconsistent date formats

Each issue was resolved with a specific rule: median imputation for age, “Unknown Customer” for blank names, de-duplication, removal of invalid rows, and a single standardized date format.



QUALITY GATE



Output: Cleaned_Sales_Dataset.csv — analysis-ready for Tasks 2 & 3.

What the Numbers Say

1,500 orders cleaned in SQL and visualized in an interactive Excel dashboard.

₹20.73M

TOTAL SALES

1,500

TOTAL ORDERS

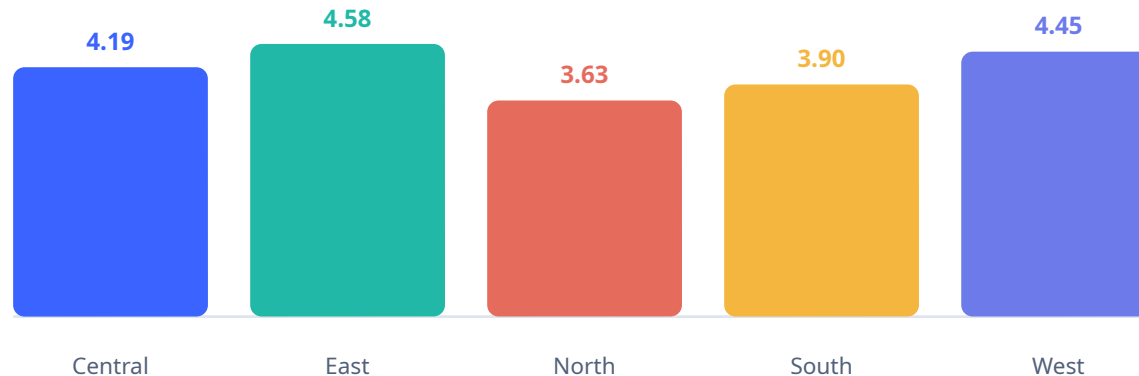
4,534

UNITS SOLD

₹3.82M

MARKETING SPEND

REVENUE BY REGION • ₹ MILLIONS



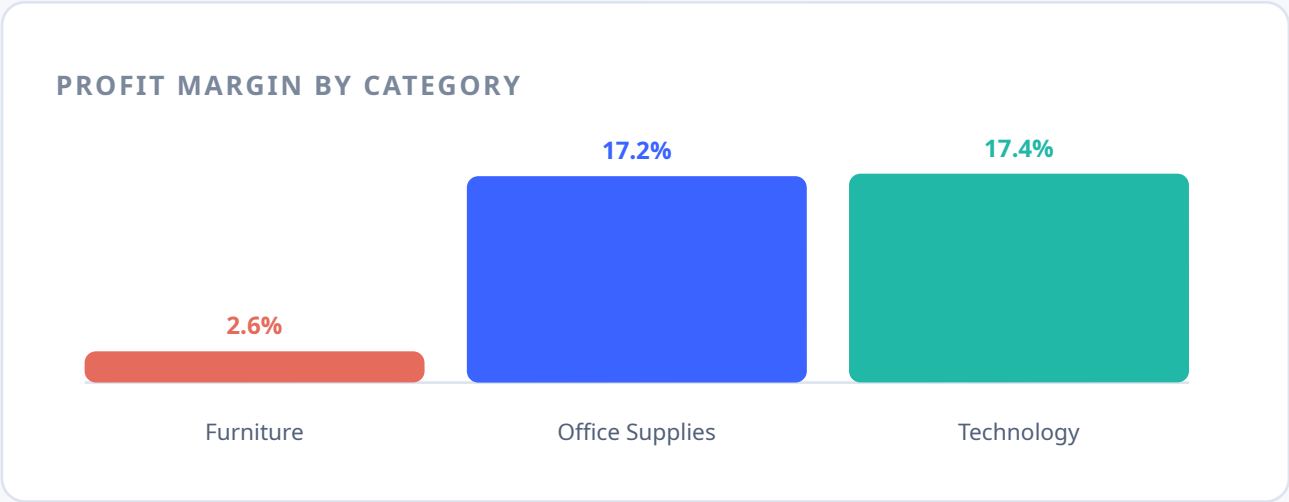
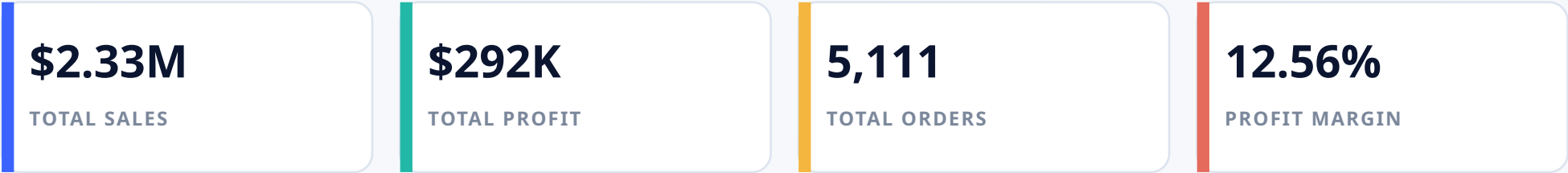
7 BUSINESS QUESTIONS

- Top revenue-generating product categories
- Monthly revenue trend
- Best-performing sales channels
- Marketing spend vs. revenue
- Regional performance gaps

INSIGHT East leads regional revenue; North trails ~21% behind — a gap worth investigating in Task 3.

Sales Volume Isn't Profit

Superstore dataset · cleaned in Python · visualized in Tableau



THE RED FLAG

Furniture generates \$754.7K in sales, almost as much as Technology (\$839.9K), but returns only \$19.7K in profit.

A large profitability gap emerges and requires statistical validation before becoming a business recommendation.



Turning an Observation Into a Testable Claim

The key question: is Furniture's lower profitability meaningfully different from the rest of the categories?

H_0 • NULL HYPOTHESIS

There is no significant difference in profit margin between the Furniture category and other product categories.

H_1 • ALTERNATIVE HYPOTHESIS

Furniture's profit margin is significantly lower than that of other product categories.

TEST SELECTED

Independent Samples T-Test (Welch's correction)

Comparing mean per-order profit margin (%) between Furniture orders and all other orders.

$\alpha = 0.05$

The Test Confirms a Significant Gap



T-STATISTIC

-11.03

P-VALUE

5.8×10^{-28}

95% CONFIDENCE INTERVAL

$[-11.95, -8.34]$

DECISION

Reject H_0

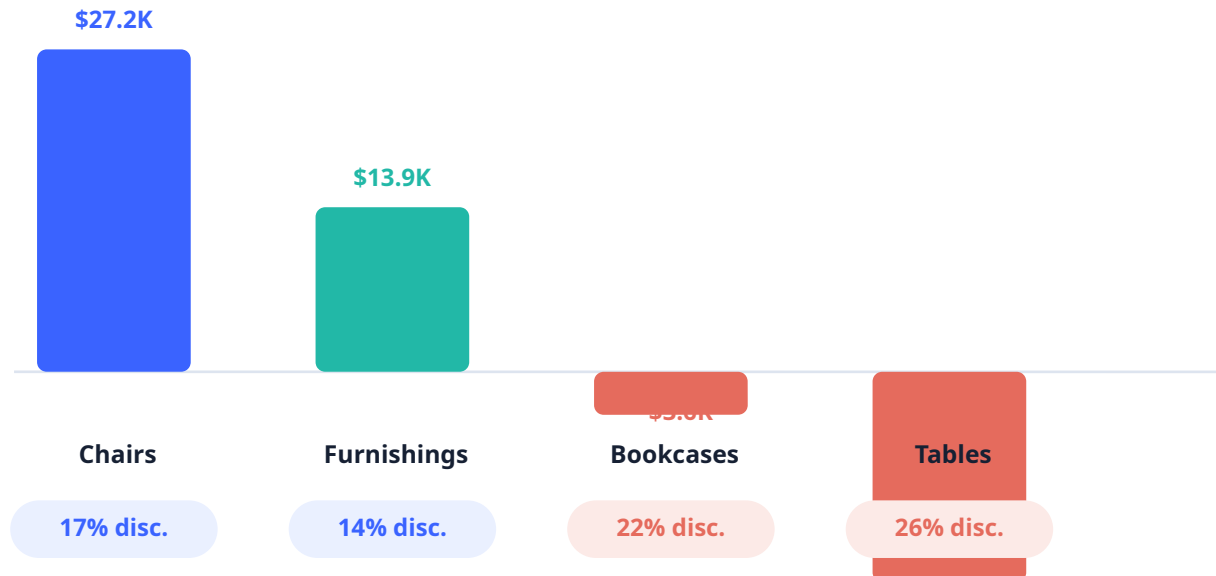
The observed gap is statistically significant, not chance.

Note: the test uses per-order averages, while the category-level margin on the prior slide is an aggregate measure. Both indicate lower Furniture profitability.

Not All Furniture Is the Problem

Profit by Furniture sub-category, paired with average discount rate.

PROFIT BY FURNITURE SUB-CATEGORY



THE PATTERN

Chairs and Furnishings are healthy, profitable product lines.

Tables and Bookcases are sold at 22–26% average discount and are net loss-making.

So the category-level problem is concentrated in two sub-categories, not the whole Furniture portfolio.

What the Business Should Do Next

01

Test a discount cap

Tables and Bookcases are net loss-making at 22–26% average discount. Pilot a lower threshold, such as 15%, and monitor margin recovery.

02

Review Tables supplier cost

A discount cap alone may not fix the ~\$17.8K loss on Tables; supplier economics deserve a separate review.

03

Redirect growth spend

Technology and Office Supplies show 17%+ margins in the analysis; evaluate whether growth spend can be shifted toward these categories.

04

Make validation quarterly

Re-run the same statistical test each quarter so margin erosion is identified early.

Principle: convert the validated finding into a monitored business action, then measure whether the intervention changes margins.

Capabilities Built Across the Internship

01

Data Cleaning & Preparation

Python, Pandas and SQL used to inspect, de-duplicate and standardize over 6,000 raw records across two datasets.

02

BI Dashboards & Visualization

Interactive Excel dashboard (Task 2) and Tableau dashboard (Task 3) built with KPI cards, filters and drill-down analysis.

03

Statistical Analysis

Hypothesis formulation, Welch's t-test, p-values and confidence intervals used to validate a business claim.

04

Data Storytelling & Communication

Findings translated into a stakeholder-ready narrative with root-cause analysis and actionable recommendations.

TOOLS

Python

Pandas

SQL

Excel

Tableau

Statistics

From Tool Usage to Analytical Thinking

The biggest shift across the internship was learning to connect technical work to a business question and then validate the conclusion.

TECHNICAL

Build a repeatable workflow: clean with Python/Pandas and SQL, then communicate through Excel and Tableau.

Repeatable

ANALYTICAL

Move beyond “what happened?” to “why did it happen?” and “is the difference statistically meaningful?”.

Better questions

BUSINESS

Turn validated findings into specific actions, assumptions and follow-up measurements rather than stopping at a dashboard.

Better questions

Next step: apply the same evidence-first workflow to new business domains and larger, less-structured datasets.

Explore the Full Project Portfolio

Four repositories document the complete internship journey from data preparation through validation.

01 Task 1 — Data Cleaning

[Open GitHub repository](#)

02 Task 2 — EDA & BI Dashboard

[Open GitHub repository](#)

03 Task 3 — Deep-Dive Dashboard

[Open GitHub repository](#)

04 Task 4 — Storytelling & Validation

[Open GitHub repository](#)

MASTER PORTFOLIO This capstone deck becomes the presentation layer for the final master GitHub portfolio.

THANK YOU

Four tasks. One analytical journey.
From raw data to validated business decisions.

-  CLEAN
-  EXPLORE
-  DEEP-DIVE
-  VALIDATE

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